How Does Chain-of-Thought Help Language Models Count? A Causal Comparison of Direct and Trace-Based Mechanisms

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Abstract

Counting all matching records in a long context requires every valid source to affect the answer exactly once. In a realistic needle-in-a-haystack audit, trace-generating modes—chain-of-thought (CoT), numbered enumeration, and bullet enumeration—are substantially more accurate than direct non-thinking answers. In a core regime with at most ten needles and 10k context tokens, the three trace modes often occupy the same high-accuracy band for reliable Qwen, Gemma, and GLM cells, whereas direct accuracy is lower. Across the full grid, accuracy generally falls and absolute count error grows as the number of needles or context length increases, although the functional form is not stable enough to claim a universal empirical law.

We ask which computational change explains this advantage. Our working hypothesis contrasts *broad retrieval* in Direct mode with *targeted retrieval plus iterative aggregation* in CoT. We organize the study as a four-link causal argument: first identify the behavioral gap; then locate and causally control count representations; next identify broad-source, matched-source, and next-source attention routes; finally test whether targeted retrieval improves unique-source coverage and the discriminability of the final count state, thereby improving exact answers. Qwen3-8B and Gemma4-E4B are co-primary mechanism models. Attention maps and PCA are used only for discovery and visualization; causal claims require query-local ablation, clean activation restoration, natural-state patching, and free-running validation. Existing synthetic results demonstrate that the proposed dissociations are measurable, while realistic LLM mechanism results remain explicitly registered experiments. **[TODO: Replace the final status sentence with paired effect sizes, cross-model state-quality estimates, and arrow-level causal recovery intervals.]**

1 Introduction

Long-context retrieval benchmarks usually ask whether a model can recover one item from distractors. Counting is stricter: every valid record must contribute once, while invalid and repeated records must contribute zero. A correct answer therefore requires source discovery, duplicate-free coverage, aggregation, stopping, and numerical readout. Multi-needle tasks such as RULER expose failures that single-needle accuracy can conceal [4].

CoT often improves multi-step reasoning [12], but the emitted trace need not faithfully report the computation that produced the answer [9]. The question of this paper is therefore mechanistic:

> *How does CoT help a language model count, and which causal computation distinguishes it from a direct non-thinking answer?*

We test a deliberately small contrast. Direct answering may retrieve many relevant sources through a broad downstream route and compress their contributions into one late total state. CoT may instead issue a sequence of source-specific retrievals, write the retrieved items into an external trace, update a progress or aggregate state, and use that state to retrieve the next source or stop. The central benefit would then be a scheduling change: a difficult high-fan-in retrieval-and-aggregation operation is replaced by repeated low-fan-in retrievals whose intermediate results remain available in the generated prefix. We do not assume in advance that either mode contains a scalar neural counter, that a visible index is faithful, or that more generated tokens are sufficient.

The paper follows one evidence chain:

$$\begin{aligned}\text{behavioral advantage} &\longrightarrow \text{count representation} \longrightarrow \text{routing circuit}\\ &\longrightarrow \text{lower functional noise} \longrightarrow \text{exact count.}\end{aligned} \tag{1}$$

Each arrow has a separate falsification test. A probe or PCA curve can establish representation, not use. An attention heatmap can nominate a route, not establish its function. An ablation can show necessity, but the proposed explanation of CoT requires restoration of the intervening retrieval or count state and recovery of the complete parsed answer.

Our contributions are:

1. We consolidate the behavioral phenomenon over needle count, context length, response mode, and a broad model panel, while treating any parametric "law" as a held-out prediction problem rather than a fitted slogan.
2. We define the minimal state variables needed to compare Direct terminal totals with CoT progress and final totals, and specify geometric steering and natural-state patching tests that distinguish a readable count code from a causally executable counter.
3. We define three attention-routing roles—broad-source, matched-source, and next-source routing—and give one common causal certification ladder for all three.
4. We register the decisive mediation experiment: perturb targeted retrieval, measure source identity and unique coverage, measure the resulting final-state discriminability, and restore the clean intermediate or final state while the upstream route remains perturbed.

The realistic mechanism study is co-primary in Qwen3-8B and Gemma4-E4B [2, 13]. The model identities, semantic sites, endpoints, and intervention logic are shared, but heads are aligned by functional role and relative depth rather than by numerical head index. Synthetic transformers serve as a controlled bridge for measurement design and training dynamics; they do not substitute for causal results in pretrained LLMs.

2 Behavioral Phenomenon and Empirical Response Surfaces

2.1 Task, response modes, and outcomes

A context x contains candidate records $R_1, \dots, R_M$. A registered predicate $v_i = \phi(R_i, x) \in \{0, 1\}$ determines whether record i is a valid counting unit, and the gold answer is

$$C(x) = \sum_{i=1}^{M} v_i. \tag{2}$$

In the current benchmark, the experimental target count is denoted by $N = C(x)$, and context length by L. Occurrence counting and unique-source counting are separate tasks; duplicates contribute according to the preregistered unit and are never silently deduplicated during scoring.

We compare four observable response modes:

1. **Direct**: return only the final count with thinking disabled;
2. **Index**: explicitly enumerate matched records in a numbered list before the count;
3. **Bullet**: explicitly enumerate matched records in an unnumbered list before the count;
4. **CoT**: use the model's native thinking mode and then return the count.

Index and Bullet are controlled forms of visible serial computation; native CoT is ecologically natural but has model-selected trace structure. The primary endpoint is numerical exactness of the parsed integer. Parse success, strict format compliance, truncation, trace precision/recall, duplicate rate, and final-versus-enumerated consistency are reported separately. This prevents an accurate number with extra text from being conflated with a counting failure and prevents a truncated trace from being treated as evidence about aggregation.

For model r, mode m, target count N, and length L, define

$$A_{rm}(N,L) = \Pr(\widehat{C} = C \mid r,m,N,L), \qquad e = \widehat{C} - C, \qquad D_{rm}(N,L) = \mathbb{E}[|e| \mid r,m,N,L]. \quad (3)$$

The paired CoT advantage is

$$\Delta_r(N,L) = A_{r,\mathrm{CoT}}(N,L) - A_{r,\mathrm{Direct}}(N,L). \quad (4)$$

All mode effects are paired within the same generated context and seed. Numerical exactness is primary; signed bias and absolute error diagnose undercounting and error magnitude.

2.2 The behavioral model panel

The empirical response-surface panel is intentionally broader than the mechanism panel. It contains a within-family Qwen size series and several cross-family reasoning models:

Table 1: Frozen behavioral panel and its role. Gemma4-E4B has approximately 4.5B effective/non-embedding parameters and 8B parameters when embeddings are included; we therefore report both conventions and do not treat it as a dense 4B point.

Role	Models	Use and restriction
Within-family size panel	Qwen3-4B, 8B, 14B, 32B	Develop candidate surfaces on 4B/8B/32B; use 14B as a fully held-out interpolation test. Do not refit the functional form after seeing 14B.
Cross-family behavior	Gemma4-E4B; GLM-4-9B and GLM-Z1-9B; Cogito-v1-Preview-8B; Nemotron-Nano-v2-9B; Nemotron-3-Nano-4B	Test direction and heterogeneity of the N, L effects. Do not pool their parameter counts into one scaling coefficient.
Co-primary mechanism	Qwen3-8B and Gemma4-E4B	Direct versus native CoT is primary; Index is a structured positive control. Roles transfer by semantic site and causal function, not head number.

The current focused analysis contains eight logical slots: Qwen3-4B/8B/32B, Gemma4-E4B, one GLM slot, Cogito-v1-Preview-8B, and the two Nemotron models. Qwen3-14B is absent and is a confirmatory TODO. The GLM slot is stitched: GLM-4-9B supplies Direct/Index/Bullet, whereas GLM-Z1-9B supplies CoT. It is useful descriptively but is not a same-weight thinking toggle. Models outside this panel are retained in the appendix with numerical accuracy and an explicit inclusion failure, rather than being silently dropped.

2.3 Preliminary core-regime result

The realistic V2 audit contains 53 model–mode cells and 500 requests per complete cell, for 26,500 requests. The grid is

$$N \in \{1, 2, 3, 4, 5, 6, 8, 10, 20, 30\}, \qquad L \in \{2, 3, 5, 10, 20\}\text{k tokens},$$

with ten context seeds per (N, L) cell. We define the *core regime* before analysis as $N \le 10$ and $L \le 10$k, giving 320 requests per complete model–mode cell.

Table 2: Preliminary numerical exact accuracy in the core regime $N \le 10, L \le 10$k (320 requests per available cell). The table reports behavior, not a same-checkpoint causal mode effect. Qwen3-14B has not yet been run.

Model / logical slot	Direct	Index	Bullet	CoT
Qwen3-4B	.519	.934	.919	.988
Qwen3-8B	.716	.991	.991	.997
Qwen3-14B	*TODO: all four modes*			
Qwen3-32B	.781	.994	.997	.994
Gemma4-E4B	.303	.966	.941	.922
GLM-4 / GLM-Z1†	.381	.853	.844	.834
Cogito-v1-Preview-8B	.506	.625	.938	.897
Nemotron-Nano-v2-9B	.322	.825	.738	.844
Nemotron-3-Nano-4B	.478	.797	.619	.794

†Direct/Index/Bullet use GLM-4-9B; CoT uses GLM-Z1-9B. The row is excluded from same-weight mode-effect estimates.

Three empirical patterns motivate the mechanism study. First, all available trace-generating cells in table 2 exceed their Direct counterpart. Second, CoT, Index, and Bullet occupy a similarly high band for the reliable Qwen, Gemma, and GLM cells; deviations such as Cogito Index and Nemotron Bullet show that trace format is not interchangeable for every model. Third, Direct is already accurate on some easy cells, which permits paired success/failure and non-ceiling mechanism analyses rather than comparing only a successful CoT state with a failed Direct state.

Across the full grid, accuracy generally decreases with larger N and L, while $|e|$ generally increases. The trend is not pointwise monotone for every model and mode. For example, parsing and truncation can distort Gemma4-E4B Direct length summaries, and the shared Direct response form performs poorly for Qwen3-8B. We therefore state the trend qualitatively until the held-out tests below are complete. The mechanism paper does not depend on a universal scaling law.

TODO. B0—paired behavioral effect to explain. For Qwen3-8B and Gemma4-E4B, rerun the same 500 registered contexts under Direct, Index, Bullet, and native CoT with frozen checkpoint and tokenizer revisions. Cross response mode with (i) deterministic matched decoding and (ii) each model's documented native policy, and cross CoT with a fixed generated-token budget. Report paired numerical exactness, absolute error, unique-source recall, duplicates, stop correctness, parse, format, and truncation with context-seed cluster-bootstrap intervals. Predeclare overlapping-difficulty cells for Gemma4-E4B because its Direct accuracy is near floor. If the numerical CoT–Direct gap disappears after decoding and budget matching, stop the reasoning-specific mediation claim.

2.4 Response surfaces are descriptive unless they predict held-out models

We use one generic family of response surfaces for both accuracy and error:

$$\text{logit}\, A_{rm}(N, L) = \alpha_{rm}^{(A)} + \beta_{N,rm}^{(A)} f(N) + \beta_{L,rm}^{(A)} g(L) + \beta_{NL,rm}^{(A)} f(N) g(L), \tag{5}$$

$$\log(1 + D_{rm}(N, L)) = \alpha_{rm}^{(D)} + \beta_{N,rm}^{(D)} f(N) + \beta_{L,rm}^{(D)} g(L) + \beta_{NL,rm}^{(D)} f(N) g(L). \tag{6}$$

Candidate bases $f(N) \in \{N, \log N\}$ and $g(L) \in \{L/1000, \log(L/1000)\}$, interactions, regularization, and held-condition splits are frozen before Qwen3-14B is generated. Coefficients are partially pooled within mode but remain model specific. A parametric form is promoted to an empirical law only if it predicts the held-out 14B results better than a no-size hierarchical baseline and remains stable under leave-one-size-out and family-held-out checks.

Current evidence does not meet that standard. The Native $N \times L$ interaction is the most stable descriptive term across the eight development slots. Direct has useful average predictive structure but unstable coefficients and a negative cross-validated R^2 for Qwen3-8B under the shared form. Index and Bullet have weak or unreliable shared surfaces. Accordingly, the main text will emphasize nonparametric (N, L) heatmaps and partial contrasts; full coefficients and failed fits belong in the appendix.

TODO. B1—held-out law test and presentation. Using only Qwen3-4B/8B/32B, freeze the candidate basis, partial-pooling model, and all splits. Generate Qwen3-14B on the identical 50 (N, L) cells and ten seeds in all four modes (2,000 requests). Report held-out log score, calibration, accuracy error, and absolute-error prediction against a no-size baseline; do not reselect after observing 14B. The main behavior figure should combine (a) Qwen size-panel predictions with the untouched 14B points and (b) Direct-versus-CoT cross-model partial-effect intervals. If validation succeeds, preregister a second extrapolation test with additional small Qwen checkpoints; if it fails, do not expand the size panel, replace "law" with qualitative response surfaces, and retain coefficients only as descriptive summaries.

TODO. B2—protocol and checkpoint audit. Either evaluate all four response modes on each of GLM-4-9B and GLM-Z1-9B or keep the stitched GLM slot descriptive-only. Repair and rerun Gemma4-E4B Direct and Cogito Index parser/protocol failures, reporting all-request numerical exactness, registered success, parsed-only bias, and a selection-sensitivity analysis side by side. Freeze a model-level inclusion code before examining mechanism activations.

2.5 Mechanism sample and mode choice

Mechanism data use a dense $N = 1{:}30$ subset at 4k tokens: ten independently generated contexts per count, for 300 prompts per model and mode. A sparse 8k replication uses $N \in \{1, 5, 10, 20, 30\}$

with ten contexts per count. Qwen3-8B and Gemma4-E4B are both analyzed; Direct and native CoT are primary, while Index is a structured serial-computation control. Bullet remains in the behavioral analysis unless it reveals a qualitatively distinct trace mechanism.

Canonical teacher-forced traces provide unambiguous source alignments and semantic sites. Free-running traces provide execution evidence. Every generated item is aligned to one source or labeled duplicate, hallucinated, or unresolved; no error category is dropped. Discovery and reporting splits are separated by document, source identities, context seeds, and, where possible, count values. Primary mode analyses include all paired prompts; both-correct, Direct-only, CoT-only, and both-wrong strata are secondary diagnostics, not conditioning criteria.

3 Counter Representations and Causal State Control

3.1 Two minimal computation graphs

The behavioral gap leaves open where the total is represented and whether a model updates it sequentially. We therefore begin with two minimal computation graphs, without assigning individual heads or neurons in advance.

Direct: broad retrieval followed by a terminal total. Let ρ_i^D be the task-relevant contribution retrieved from record R_i. The Direct candidate is

$$\rho_i^D = \mathcal{R}_D(x, R_i, v_i; q_D), \qquad Z_D = G_D(\rho_1^D, \ldots, \rho_M^D), \qquad \widehat{C}_D = Q_D(Z_D). \tag{7}$$

Here q_D is the position immediately before the first answer digit. "Broad" means that multiple valid records have independently measurable causal effects through a common downstream route or route bundle. It does not require uniform attention or one globally broad head. "Terminal" means that the causally sufficient total Z_D is constructed near the answer; it does not deny that earlier prompt states contain partial information.

CoT: targeted retrieval followed by iterative aggregation. At trace step k, let π_k identify the intended source, e_k the written item, and Γ_k the aggregate carrier formed by an optional internal state together with the visible trace and its KV cache. The CoT candidate is

$$\pi_k = \Pi_C(x, \Gamma_{k-1}), \qquad e_k = W_C\Big(\mathcal{R}_C(x, R_{\pi_k}; q_k^{\mathrm{ret}})\Big), \tag{8}$$

$$\Gamma_k = F_C(\Gamma_{k-1}, e_k), \qquad K = \min\{k : \sigma_C(\Gamma_k) = \textsc{stop}\}, \tag{9}$$

$$Z_C = G_C(\Gamma_K), \qquad \widehat{C}_C = Q_C(Z_C). \tag{10}$$

The retrieval query q_k^{ret} occurs immediately before the source-bearing content of item k; q_k^{end} is the item end before the next-item or close decision; q_C is immediately before the first final-answer digit. "Targeted" means that the causal source contribution at step k is concentrated on R_{π_k} and retargets under a counterfactual change of π_k. "Iterative aggregation" means that changing a semantically new, duplicate, or invalid item changes the later carrier, next retrieval, stopping decision, or final total in the predicted way. The visible trace may be the memory; an internal scalar counter is not assumed.

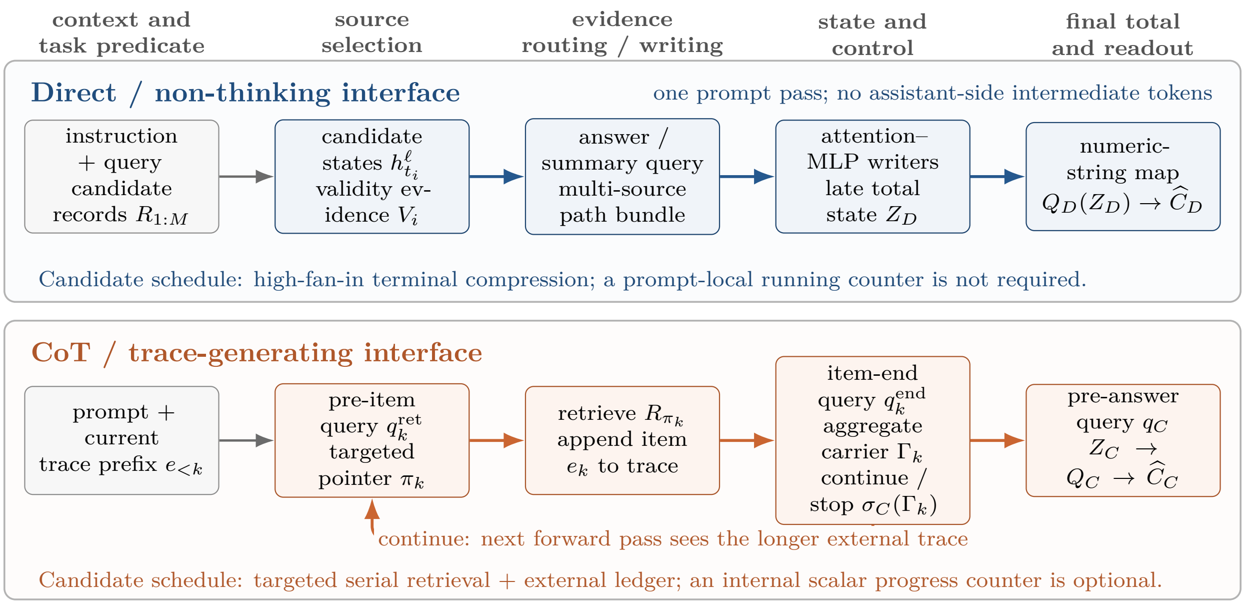

Figure 1: The two candidate computation graphs. Direct mode must make a set of sources jointly influence a late total state. CoT decomposes this operation into matched-source retrieval, trace writing, aggregate-carrier update, and next-source or stop control before final readout. Boxes are functional variables, not claims about one head or neuron.

3.2 Registered count states

At layer ℓ, we register only three residual-stream states:

$$Z^\ell_D = h^\ell(q_D), \qquad P^\ell_{C,k} = h^\ell(q^{\mathrm{end}}_k), \qquad Z^\ell_C = h^\ell(q_C). \tag{11}$$

Z^ℓ_D and Z^ℓ_C are candidate *final-total representations* of the same variable C. $P^\ell_{C,k}$ is a candidate *progress state* whose semantic label is the number k_{unique} of distinct valid sources covered so far, not the emitted item number or the last visible numeral. State quality is compared across modes only at the matched final-total sites Z^ℓ_D and Z^ℓ_C; a Direct total is never compared with a CoT progress state.

This distinction resolves where to search for a non-thinking counter. We first test for a total code at q_D across prompts with different C. We then scan states after candidate records and matched filler boundaries. A prompt-side state is called an *iterative counter* only if one new valid record causes a +1 update, an invalid or already-counted record causes a no-op, the update persists, and transplanting it changes the final count. If these event-level tests fail while Z^ℓ_D remains causally transportable, the Direct mechanism contains a terminal total representation but no demonstrated prompt-side counter. Cross-prompt decoding at q_D alone cannot answer this question.

For CoT, let

$$u_k = \begin{cases} 1, & e_k \text{ covers a new valid source,} \\ 0, & e_k \text{ is invalid or duplicates a covered source.} \end{cases}$$

A component of $P^\ell_{C,k}$ is an iterative counter only if a held-out semantic readout g satisfies

$$g(P^\ell_{C,k}) - g(P^\ell_{C,k-1}) \approx u_k \tag{12}$$

and intervention on that component changes a downstream retrieval, stopping decision, or final count. If only trace-token or KV-cache patches transport progress, the supported mechanism is an external trace ledger rather than an internal residual counter.

3.3 Counter curves and representation quality

Linear decoding asks whether count information is present; it does not describe geometry or causal use. Let z collect preregistered nuisances such as context length, answer position, target-depth statistics, trace length, and local token identity. Residualize the registered state using a fit split,

$$\widetilde{h}_s^\ell = h_s^\ell - \widehat{\mathbb{E}}[h_s^\ell \mid z],$$

and define the held-out count centroid

$$\mu_s^\ell(c) = \mathbb{E}[\widetilde{h}_s^\ell \mid C = c]. \tag{13}$$

The *counter curve* at site s and layer ℓ is the ordered sequence of full-space centroids together with adjacent displacements

$$\mathcal{K}_s^\ell = \left(\mu_s^\ell(c_1), \dots, \mu_s^\ell(c_J); \delta_j^\ell = \mu_s^\ell(c_{j+1}) - \mu_s^\ell(c_j)\right). \tag{14}$$

A two-dimensional PCA plot is only $\Pi_2\mathcal{K}_s^\ell$: the basis is fitted on the discovery split and held fixed for reporting data. Ordering or visual smoothness in PCA is descriptive and does not show that the model traverses the curve during computation.

We use "noise" only as shorthand for conditional representation dispersion across matched prompts, not stochasticity of a deterministic hidden state. For adjacent observed counts, define

$$t_j^\ell = \frac{\mu^\ell(c_{j+1}) - \mu^\ell(c_j)}{\|\mu^\ell(c_{j+1}) - \mu^\ell(c_j)\|_2}, \qquad d_j^\ell = \frac{\|\mu^\ell(c_{j+1}) - \mu^\ell(c_j)\|_2}{c_{j+1} - c_j}, \tag{15}$$

$$\overline{\Sigma}_j^\ell = \frac{\mathrm{Cov}(\widetilde{h}^\ell \mid c_j) + \mathrm{Cov}(\widetilde{h}^\ell \mid c_{j+1})}{2}, \qquad \mathrm{SNR}_j^\ell = \frac{(d_j^\ell)^2}{(t_j^\ell)^\top \overline{\Sigma}_j^\ell t_j^\ell + \epsilon}. \tag{16}$$

SNR_j^ℓ compares unit-count centroid spacing with within-count dispersion in the local count direction. We report the two terms separately, because a low SNR may arise from smaller spacing, larger dispersion, or both. Normal-direction dispersion, shrinkage Mahalanobis distance, decoder bias/variance, and nuisance sensitivity are appendix analyses.

TODO Figure: Counter representation and causal control

Layerwise readability	*Held-out curves*	*Causal transport*
Qwen3-8B / Gemma4-E4B	Direct Z_D / CoT $P_{C,k}, Z_C$	steering / natural patching
decoder error by layer	PCA with uncertainty	parsed-count dose response
SNR by count and mode	full-space spacing	random/orthogonal controls

Use the same model colors across panels; Direct and CoT must be compared at matched final-total sites. PCA is visualization only.

Figure 2: Planned representation figure. The left and middle panels establish where count and progress are represented; the right panel tests whether movement of that state controls the complete parsed count.

TODO. R1—counter-curve map. For each co-primary model and Direct/CoT mode, collect the dense 4k mechanism set and sparse 8k replication. At every layer, extract Z_D^ℓ, $P_{C,k}^\ell$, and Z_C^ℓ; fit nuisance regressions, PCA bases, and linear decoders on discovery documents only; evaluate on held-out documents, identities, positions, and counts. Report decoder MSE, explained PCA variance, full-space centroid spacing, tangential SNR with bootstrap intervals, and index-free/permuted-index controls. The primary fair comparison is $Z_D^\ell(C)$ versus $Z_C^\ell(C)$. Support for "larger counts are noisier" requires a preregistered negative SNR slope or positive normalized-dispersion slope after L, target depth, answer position, and trace extent are controlled; the PCA picture is not an endpoint.

3.4 Geometric steering and natural-state patching

To test whether the counter curve is executable, preserve the receiver-specific residual while moving only its count coordinate. For a receiver with count c_r, residual $\epsilon_r = h_r - \mu(c_r)$, and a point $\gamma_{r\to d}(\alpha)$ on the piecewise path through observed centroids toward donor count c_d, intervene with

$$h'_r(\alpha) = \epsilon_r + \gamma_{r\to d}(\alpha), \qquad 0 \le \alpha \le 1. \tag{17}$$

We compare this local path with the endpoint chord, norm-matched tangent-orthogonal directions, random directions, and full centroid replacement. The output endpoint is the complete parsed-count distribution or its expected value

$$\mathcal{C}(h) = \sum_c c\, p(\widehat{C} = c \mid h),$$

not the logit of the first digit.

A stronger test uses same-site natural donor states. For donor count c_d and receiver count c_r, patch the donor residual or the minimal localized state slice into the receiver and measure the transport slope

$$\beta_{\text{patch}} = \frac{\mathbb{E}[\mathcal{C}(h_{\text{patched}}) - \mathcal{C}(h_{\text{receiver}})]}{c_d - c_r}. \tag{18}$$

Bidirectional slopes near one, together with low off-manifold distance and ineffective matched random patches, support a causally sufficient total coordinate. Steering alone is weaker because an artificial direction may bypass the natural computation.

TODO. R2—geometry causality. Use disjoint state-fit and intervention prompts. For Z_D^ℓ, $P_{C,k}^\ell$, and Z_C^ℓ, run bidirectional adjacent-count steering, non-adjacent curve-versus-chord steering, residual-preserving centroid movement, and same-position natural donor transplants. Balance $\delta = \pm 1, \pm 2$, length, answer position, trace extent, and correctness stratum. Report transport slope, path-tracking error, adoption of the donor parsed count, activation distance to natural states, downstream KL, and complete free-running exactness. A causally executable count state requires a monotone bidirectional dose response, superiority to norm-matched orthogonal/random controls, and agreement between geometric and natural-state interventions.

TODO. R3—iterative counter versus terminal total. For Direct, create fixed-token-length record triples in which one candidate is new-valid, invalid, or a duplicate while candidate ordinal and local depth are fixed. Test the immediate record-end state, subsequent matched filler boundaries, and q_D. For CoT, create item-prefix triples with the same visible item number and length but new-valid, invalid, or duplicate semantics; patch residual state and trace-token/KV state separately at q_k^{end}. Require +1, no-op, persistence, held-out identity generalization, and a downstream retrieval/stop/final-answer effect before using "iterative counter." q_D/q_C-only transport supports a final-total representation; trace/KV-only transport supports an external ledger.

4 Routing Heads and Counting Circuits

4.1 Three routing roles

The state analysis identifies what is represented; it does not identify how source information reaches that state. We use attention only to rank candidates on a discovery split, then assign a functional name only after query-local causal tests.

Let $A_{qj}^{\ell h}$ be attention from semantic query q to key token j in layer ℓ, query head h. For source span S_i, define its attention mass

$$a_i^{\ell h}(q) = \sum_{j \in S_i} A_{qj}^{\ell h}. \tag{19}$$

All source scores are averaged within an example before averaging across examples, so long traces do not dominate. Exact spans, query rows, and missing/alignment rules are frozen before head discovery.

Broad-source routing score. At the Direct pre-answer query q_D, let

$$\mathcal{M}(q_D) = \sum_{i=1}^{N} a_i(q_D), \qquad r_i(q_D) = \frac{a_i(q_D)}{\mathcal{M}(q_D) + \epsilon},$$

and define normalized source entropy

$$H_N(q_D) = -\frac{\sum_{i=1}^{N} r_i(q_D) \log(r_i(q_D) + \epsilon)}{\log N}.$$

The discovery score is

$$B_{\ell h} = \mathbb{E}[\mathcal{M}(q_D) H_N(q_D)]. \tag{20}$$

$B_{\ell h}$ is high only when a head places non-negligible total mass over multiple valid sources. We also report $\mathcal{M}$, H_N, source coverage, and the union coverage of top-k head bundles. B is not used for $N < 2$.

Matched-source routing score. At the CoT retrieval query q_k^{ret}, the intended source is S_{π_k}. The primary ranking statistic is raw matched-source mass

$$T_{\ell h}^{\mathrm{raw}} = \mathbb{E}_x \left[\frac{1}{K_x} \sum_{k=1}^{K_x} a_{\pi_k}^{\ell h}(q_k^{\mathrm{ret}}) \right]. \tag{21}$$

We additionally report conditional selectivity

$$T_{\ell h}^{\mathrm{sel}} = \mathbb{E}_{x,k} \left[\frac{a_{\pi_k}^{\ell h}(q_k^{\mathrm{ret}})}{\sum_i a_i^{\ell h}(q_k^{\mathrm{ret}}) + \epsilon} \right], \tag{22}$$

source-subset top-1, and the correct-versus-best-wrong QK margin. Raw mass prevents a head with negligible source attention from ranking highly merely because its tiny mass is conditionally selective.

Next-source routing score. At item end q_k^{end}, for a canonical ordering with a unique next unvisited source, define

$$S_{\ell h}^{\mathrm{next}} = \mathbb{E}_x \left[\frac{1}{K_x - 1} \sum_{k<K_x} a_{\pi_{k+1}}^{\ell h}(q_k^{\mathrm{end}}) \right]. \tag{23}$$

We call a high-scoring component a *next-source routing candidate*, not a successor head. The stronger term *continue/stop control head* is reserved for a component whose bidirectional patch changes both the next source and whether generation continues. Native traces may not have a unique gold order; canonical traces provide the primary score, while aligned free-running traces validate natural execution. The role is distinct from classic successor heads that implement an ordered-token map such as $k \mapsto k+1$ [3].

Table 3: The three headline routing roles. A score nominates candidates; the final column is required for causal naming.

Role	Query	Registered keys	Score	Causal certification
Direct broad-source	q_D	all valid prompt sources	$B_{\ell h}$	multiple source-specific effects pass through one bundle; local ablation damages Z_D and the answer; clean patch restores both
CoT matched-source	q_k^{ret}	intended source S_{π_k}	$T_{\ell h}^{\mathrm{raw}}$	same-position identity corruption is repaired, and counterfactual pointer changes retarget the same bundle
CoT next-source	q_k^{end}	next unvisited source	$S_{\ell h}^{\mathrm{next}}$	short/long-prefix patches change next-source and continue/close decisions in both directions and alter rollout stopping

Final trace-to-answer attention is analyzed as a readout route in the appendix, not as a fourth headline mechanism. High trace mass does not imply that a head transports a scalar total; final-total transport is already tested at Z_C in section 3.

TODO Figure: Routing roles and causal certification

Broad-source at q_D	*Matched-source*	*Next-source*
Qwen: source-wide heatmap	Qwen: step–source diagonal	Qwen: item to next source
Gemma: source-wide heatmap	Gemma: step–source diagonal	Gemma: item to next source

Under each heatmap: discovery score, query-local ablation damage, clean-patch recovery, and matched-random interval. Show source spans rather than full-token matrices, use one shared scale within each role, and mark the intervened query row.

Figure 3: Planned attention-and-causality figure. Heatmaps describe candidate routing; the aligned ablation and restoration panels determine function. Head numbers need not match across architectures.

4.2 One causal certification ladder

Each candidate role uses the same evidence ladder:

$$\text{attention signature} \longrightarrow \text{query-local necessity} \longrightarrow \text{clean restoration} \longrightarrow \text{free-running effect.} \quad (24)$$

For a clean/corrupt endpoint m, normalized recovery is

$$\mathrm{Rec} = \frac{m_{\text{patch}} - m_{\text{corrupt}}}{m_{\text{clean}} - m_{\text{corrupt}}}. \quad (25)$$

We report the raw numerator and denominator and compute recovery only when the clean–corrupt gap exceeds a frozen minimum. Endpoints are complete source-identity margins, continue-versus-close margins, final-state metrics, or complete numeric-string probabilities; no mechanism conclusion relies on one next-token logit.

Interventions proceed from coarse to fine: residual/block output, attention output versus MLP output, and then query-local pre-output-projection head slices. This ordering matters for grouped-query attention in Qwen and Gemma: shared key/value heads do not make a numerical query-head index a portable functional unit. Pattern, source-value, pre-o_{proj} head output, and post-layer residual patches are separated. Mean replacement at the registered query is primary; zero and global ablations are sensitivity analyses.

4.3 Direct broad-source circuit

The Direct experiment uses length-matched source edits that add, delete, or validity-flip one record while holding candidate ordinal, syntax, and depth fixed. A broad-source bundle must carry independent effects from multiple sources into Z_D. A set of narrow heads with broad union coverage is allowed; the claim is about the route bundle, not the visual breadth of one head. Multi-hop routes through prompt summary tokens are also tested, because absence of direct answer-to-source attention does not imply absence of broad retrieval.

TODO. H1-D—Direct broad-source route. Split each model's 300 dense prompts into discovery and reporting sets balanced by count. On discovery data, rank single heads and cumulative bundles by $B_{\ell h}$, total source mass, and union coverage. On reporting data, apply fixed-length source addition/deletion, correct-identity/wrong-attribute, and matched non-target edits. At q_D, run query-local mean ablation and clean-to-corrupt restoration for attention pattern, source value, pre-o_{proj} head output, attention output, and residual state. Primary endpoints are the number of distinct valid sources whose effects pass through the bundle, change/recovery of the frozen Z_D total-state margin, and change/recovery of the complete parsed-count distribution. Compare with same-layer score-matched random bundles, wrong query rows, wrong source spans, and candidate summary-token paths. Assign "causal broad-source route" only if all three endpoints replicate in both co-primary models.

4.4 CoT matched-source circuit

The matched-source experiment changes the identity of the intended source while preserving total count, trace step, source position, and token length. A true targeted route must follow semantic source identity rather than the visible index or absolute position. We therefore use index-free and permuted-index traces and source swaps that keep the answer unchanged.

TODO. H1-C—CoT matched-source route. Construct at least 240 same-count, same-position, same-length clean/corrupt pairs per model. Change π_k while preserving all other source locations. Select the top-K bundle by T^{raw}, conditional selectivity, and QK margin on discovery data; freeze K before reporting. At q_k^{ret}, ablate and patch attention pattern, matched-source value, pre-o_{proj} head output, attention output, and residual state. Primary endpoints are correct-versus-best-wrong source margin, post-retrieval source-identity decoding, and generated item identity. Wrong-source donors, wrong query rows, same-layer random bundles, index permutations, and non-source tokens are controls. The role is certified only if clean patches recover identity and the same bundle retargets under a pointer/source swap. Repeat on free-running prefixes before the first trace error.

4.5 Next-source and continuation control

A high S^{next} score may reflect attention to a likely future source without controlling it. We therefore construct nested prefixes that share the same current item but differ in whether another source remains. This creates both continue and terminal decisions at the same semantic and positional anchor.

TODO. H2—next-source and continue/stop control. Using canonical traces, construct at least 240 nested short/long prefix pairs with the same current item, visible index, token length, and item-end position but different remaining-source sets. At q_k^{end}, patch the discovery-ranked next-source bundle from continue to terminal examples and in the reverse direction. Measure next-source QK/source margin, generated next-item identity, continue-versus-close margin, and free-running termination location. Separately patch $P_{C,k}$, trace-token KV states, and the candidate head output to determine whether control comes from an internal progress register or external ledger. Same-layer random heads, wrong remaining sets, wrong rows, and equal-length padding are controls. Use "continue/stop control head" only after both directions and both co-primary models pass; otherwise retain "next-source routing candidate."

4.6 Cross-model role alignment

Qwen3-8B is used for discovery of the analysis workflow, not for fixing Gemma head numbers. Before inspecting Gemma effects, we freeze semantic queries, source spans, ranking metrics, bundle-size selection, causal endpoints, and negative controls. Gemma candidates are then discovered independently and evaluated on a disjoint split. A cross-model replication means the same functional role at a comparable relative-depth band produces the same causal endpoint; it does not require homologous indices or identical attention pictures.

5 From Targeted Retrieval to Lower Noise and Higher Accuracy

5.1 The proposed explanatory chain

The preceding sections can establish that count states and routing roles exist. They do not yet explain why CoT is more accurate. The explanatory hypothesis is the following causal chain:

$$H_{\mathrm{tgt}}, H_{\mathrm{next}} \longrightarrow I_{1:K}, U_{\mathrm{unique}} \longrightarrow Z_C \longrightarrow \widehat{C}. \tag{26}$$

H_{tgt} and H_{next} are the causally certified matched-source and next-source bundles; I_k is the retrieved source identity at trace step k; U_{unique} is the set or number of distinct valid sources covered; Z_C is the final-total state; and $\widehat{C}$ is the parsed answer.

The contrast with Direct mode is computational rather than merely visual. Direct must allow many source contributions to reach a terminal state before any assistant-side memory exists. As N and L grow, source coverage, contribution magnitude, and aggregation may interfere. CoT turns this high-fan-in operation into repeated matched-source queries, each of which can retrieve one source with a large correct-versus-wrong margin. Writing the result into the trace creates persistent memory for duplicate avoidance and stopping. The hypothesis predicts that CoT reduces source-identity and coverage errors and produces a better-resolved final-total state. It does not predict immunity to scale: with enough steps, retrieval, duplication, and stopping errors can accumulate, so CoT accuracy can still decrease with N and L.

5.2 Retrieval noise and count-state noise

We distinguish descriptive attention dispersion from functional retrieval failure. For a matched-source candidate, normalize attention over registered source spans:

$$p_i^{\ell h}(q_k) = \frac{a_i^{\ell h}(q_k)}{\sum_j a_j^{\ell h}(q_k) + \epsilon}. \tag{27}$$

Two attention-level diagnostics are

$$\nu_{\text{route}}^{\ell h}(k) = -\log p_{\pi_k}^{\ell h}(q_k), \qquad \nu_{\text{abs}}^{\ell h}(k) = 1 - a_{\pi_k}^{\ell h}(q_k). \tag{28}$$

The first measures conditional confusion among sources; the second detects cases in which selectivity is high only because total source mass is negligible. These quantities nominate candidate routes but are not themselves the mechanistic noise endpoint.

The primary retrieval endpoint is functional source identity. Let g_k^ℓ be the post-retrieval state and let $p_\phi(i \mid g_k^\ell)$ be a grouped held-out source-identity decoder. Define

$$\nu_{\text{id}}^\ell = \mathbb{E}[-\log p_\phi(\pi_k \mid g_k^\ell)]. \tag{29}$$

In free-running traces we additionally measure wrong-source rate, duplicate-source rate, hallucinated/unresolved rate, and unique-valid recall. A targeted route is precise only if its intervention changes these functional endpoints, not merely its attention diagonal.

For final count states, use the matched-site SNR in equation (16) and define normalized count-state dispersion

$$\nu_{\text{state,j}}^\ell = \frac{1}{\text{SNR}_j^\ell} = \frac{(t_j^\ell)^\top \overline{\Sigma}_j^\ell t_j^\ell}{(d_j^\ell)^2 + \epsilon}. \tag{30}$$

This is conditional representation dispersion per unit count separation. The primary mode comparison is Direct Z_D versus CoT Z_C; the primary intervention comparison is clean CoT versus CoT with the targeted route perturbed.

5.3 Intervening on the full chain

Let H^+ denote the clean targeted bundle and H^- a query-local ablated or semantically corrupted bundle. The mechanism predicts three linked effects:

1. H^- worsens source identity and lowers unique-valid coverage;
2. the same intervention worsens the true-count margin or SNR of Z_C;
3. while H^- remains in place, restoring the clean post-retrieval state or clean Z_C recovers the downstream count.

The first two raw effects are

$$\Delta_{\text{acc}}^T = A(H^+) - A(H^-), \qquad \Delta_{\text{SNR}}^T = \log \text{SNR}(Z_C \mid H^+) - \log \text{SNR}(Z_C \mid H^-). \tag{31}$$

For a continuous complete-count margin m_Y, final-state restoration recovery is

$$\eta_Z = \frac{m_Y(H^-, Z_C \leftarrow Z_C^+) - m_Y(H^-, Z_C^-)}{m_Y(H^+, Z_C^+) - m_Y(H^-, Z_C^-)}. \tag{32}$$

We always report all three raw margins and use η_Z only above a frozen denominator gate. The analogous η_I restores the clean post-retrieval identity state while leaving the heads perturbed. Full parsed-count probability and free-running numerical exactness are primary; teacher-forced first-token logits are diagnostics.

TODO Figure: Causal chain from retrieval to accuracy

$\boxed{\text{targeted bundle}} \rightarrow \boxed{\text{source identity / unique coverage}} \rightarrow \boxed{\text{final count-state SNR}} \rightarrow \boxed{\text{exact count}}$

Condition	Identity noise	Unique recall	Z_C SNR	Exact accuracy
Clean	low	high	high	high
Targeted ablation	↑	↓	↓	↓
Ablation + clean identity patch	rescued	rescued	partial/full rescue	partial/full rescue
Ablation + clean Z_C patch	unchanged	unchanged	restored	restored

Show paired raw effects and bootstrap intervals for Qwen3-8B and Gemma4-E4B; do not present only normalized mediation ratios.

Figure 4: Planned decisive mechanism figure. State restoration while the upstream route remains perturbed separates causal mediation from correlation between attention, representation, and accuracy.

TODO. M1—retrieval-to-state-to-answer intervention. On at least 300 paired canonical traces per co-primary model, cross targeted-bundle state (clean, query-local mean ablation, wrong-source patch) with post-retrieval identity state (natural clean versus receiver) and final-total state Z_C (natural clean versus receiver). Measure attention diagnostics, source-identity margin and ν_{id}, wrong/duplicate/hallucination rates, unique-valid recall, Z_C local SNR and true-count margin, complete parsed-count probability, and free-running exactness. Use discovery-ranked bundles frozen before the factorial experiment. Report raw path effects, η_I, η_Z, document/seed cluster-bootstrap intervals, and matched random-head, wrong-row, wrong-source, wrong-layer, and norm controls. The targeted-retrieval explanation requires the ordered degradation and rescue in equation (26); a local attention effect with no unique-coverage, state, or answer effect is a failed mechanism link.

5.4 Coverage rescue versus aggregation-noise reduction

Targeted retrieval may improve accuracy only because it finds more distinct sources. In that case, lower final-state dispersion is downstream of better coverage rather than a separate improvement in aggregation. To distinguish the accounts, we evaluate states under fixed semantic coverage.

Canonical teacher forcing supplies the same correct set of U unique sources in every condition while varying whether retrieval is naturally targeted, head-perturbed, or replaced by an equal-length prefilled ledger. If targeted-head ablation no longer changes Z_C once identity and coverage are fixed, the correct conclusion is *coverage rescue*. If Z_C SNR and the final answer still worsen at fixed coverage, and restoring the clean aggregate state rescues the count, the data support an additional *aggregation-noise reduction* effect. Both outcomes explain part of CoT, but they are not the same mechanism.

TODO. M2—fixed-coverage and extra-compute controls. Cross unique-valid coverage U, duplicate count, raw trace length, last visible index, and answer position. Compare a source-faithful CoT trace, a length-preserving one-item semantic corruption, a shuffled faithful trace, an equal-length null scratchpad, and Direct with the same faithful list prefilled in the prompt or assistant prefix. Under teacher forcing, hold the covered source set exactly fixed while ablating the targeted route; under free generation, measure how the route changes coverage. If the route affects accuracy only through U, report coverage rescue. Claim reduced aggregation noise beyond coverage only if the fixed-U intervention changes Z_C SNR/true-count margin and a clean Z_C patch restores the answer. If null scratchpads match the gain, prefer an extra-compute account.

5.5 Explaining the Direct–CoT gap

The final experiment asks whether the certified CoT chain accounts for the paired behavioral advantage in equation (4). For cells with a frozen nonzero continuous Direct–CoT margin, define operational gap removal

$$\text{GapRemoved} = \frac{m_{\text{CoT}} - m_{\text{CoT,perturbed}}}{m_{\text{CoT}} - m_{\text{Direct}}}. \tag{33}$$

The ratio is secondary and undefined near a zero denominator; paired numerical accuracy and raw margins remain primary. Because a generated trace is post-treatment, we call this operational gap removal rather than nonparametric causal mediation.

TODO. M3—cross-model gap removal and generated-prefix validation. Freeze all endpoints and intervention doses after Qwen3-8B discovery, then rerun role discovery and the full M1/M2 reporting protocol independently in Gemma4-E4B. Use all paired prompts plus preregistered overlapping-difficulty cells; common-correct subsets are sensitivities only. On 300 free-running traces per model, evaluate the frozen source-identity, progress-update, next-source/stop, Z_C, and answer effects before, at, and after the first trace error. The main claim requires: (i) targeted/next-route perturbation reduces the CoT–Direct gap more than random-route and null-compute controls; (ii) clean intermediate or final-state restoration recovers a preregistered fraction of the raw gap; and (iii) the direction replicates in both models. If Gemma Direct remains at floor, report Gemma as a failure-mechanism replication rather than as unbiased cross-model mediation.

5.6 Claim ladder and falsifiers

Table 4: Evidence required for each level of the paper's claim. Passing an earlier level does not license a later conclusion.

Claim level	Required evidence	Principal falsifier or weaker conclusion
Representation	grouped held-out count/progress decoding, full-space geometry, nuisance controls	only visible index/position decodes; no generalization
Causal state	bidirectional local steering and natural-state transport change the complete count	probe/PCA changes without parsed-answer transport
Routing role	query-local damage, clean patch recovery, semantic retargeting, free-running effect	attention heatmap changes but source identity or decision does not
CoT explanation	targeted disruption worsens coverage/state/accuracy; state restoration rescues; null compute does not	null scratchpad matches gain, fixed-coverage state is unchanged, or restoration fails

6 Supplementary Synthetic Bridge and Training Dynamics

Synthetic models permit exact source alignment, dense checkpoints, and interventions that are expensive in LLMs. Their role is to develop measurements, expose confounds, and generate predictions. They do not establish that pretrained Qwen or Gemma models use the same circuit.

6.1 What synthetic v10 contributes

Synthetic v10 trains separate 4-layer, 4-head, $d_{\text{model}} = 256$ causal transformers on counts 1–30 with learned absolute positions. Within that setting, it gives five concrete design lessons:

1. final count is linearly readable in both modes, but the centroid path bends with depth, so local or piecewise steering is preferable to one global +1 direction;
2. Direct answer queries contain broad-source routing candidates, while trace queries contain matched-source bundles;
3. progress/termination states and final-total states are causally dissociable;
4. natural final-state transplants can transport donor count, whereas some trace-marker states control early close without transporting the final scalar;
5. high trace-readout attention mass does not guarantee that the selected head output carries a total.

These findings motivate the counter-curve, head-role, and mediation experiments above. They also justify the strict separation between representation, routing, and causal transport.

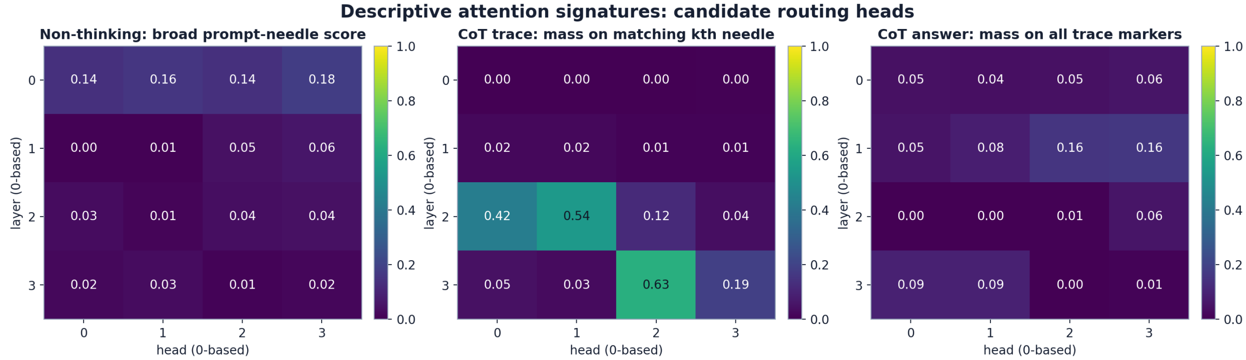

Figure 5: Preliminary synthetic v10 signatures used to design the LLM study. Direct answer queries show broad source coverage; trace queries show matched-source retrieval; trace-to-answer readout is separate. The panels are descriptive until accompanied by query-local ablation and clean patch recovery.

6.2 Why v15 is not the final mechanism setting

Version 15 uses Tiny Shakespeare backgrounds, $d_{\text{model}} = 256$, RoPE or learned relative bias, all-sequence loss, and inserted marker targets. It changes several factors at once and is therefore a design stress test rather than a clean ablation. Final counts can approach ceiling even while whole-trace fidelity falls sharply, and position-only controls at trace/final sites can be highly predictive. The lesson is negative but important: correct final answers do not prove faithful iterative aggregation, and a strong progress probe may decode position or visible step.

We therefore retain v15 only for control development: index permutation, position matching, trace-fidelity decomposition, and separation of progress from final total. It is not used as the primary synthetic claim.

6.3 v20 as the candidate training-dynamics setting

Version 20 counts occurrences of a queried set of three natural characters in a 256-character Tiny Shakespeare window, restricted to totals 1–30. It uses query-first input, RoPE, atomic count tokens, paired data/order for separately trained Direct and trace models, and a two-stage objective: all-sequence loss through step 1500, then task-output loss through step 10,000. Both models have four layers, four heads, and $d_{\text{model}} = 256$.

At the final one-seed checkpoint, the trace model reaches .912 exact count on held-out free-running examples, while the Direct model reaches .335. Whole-trace exactness and ordered-marker accuracy are also high in the trace model. The Direct output error does not exhibit a simple monotone increase with count in that sample, so the proposed count-state dispersion analysis remains necessary. Local repair artifacts suggest strong next-source and matched-source patching, including value-dominant transport, but the original run manifest marks part of the causal stage as failed. These results are hypotheses until the stage is rerun from a recorded revision with multiple seeds.

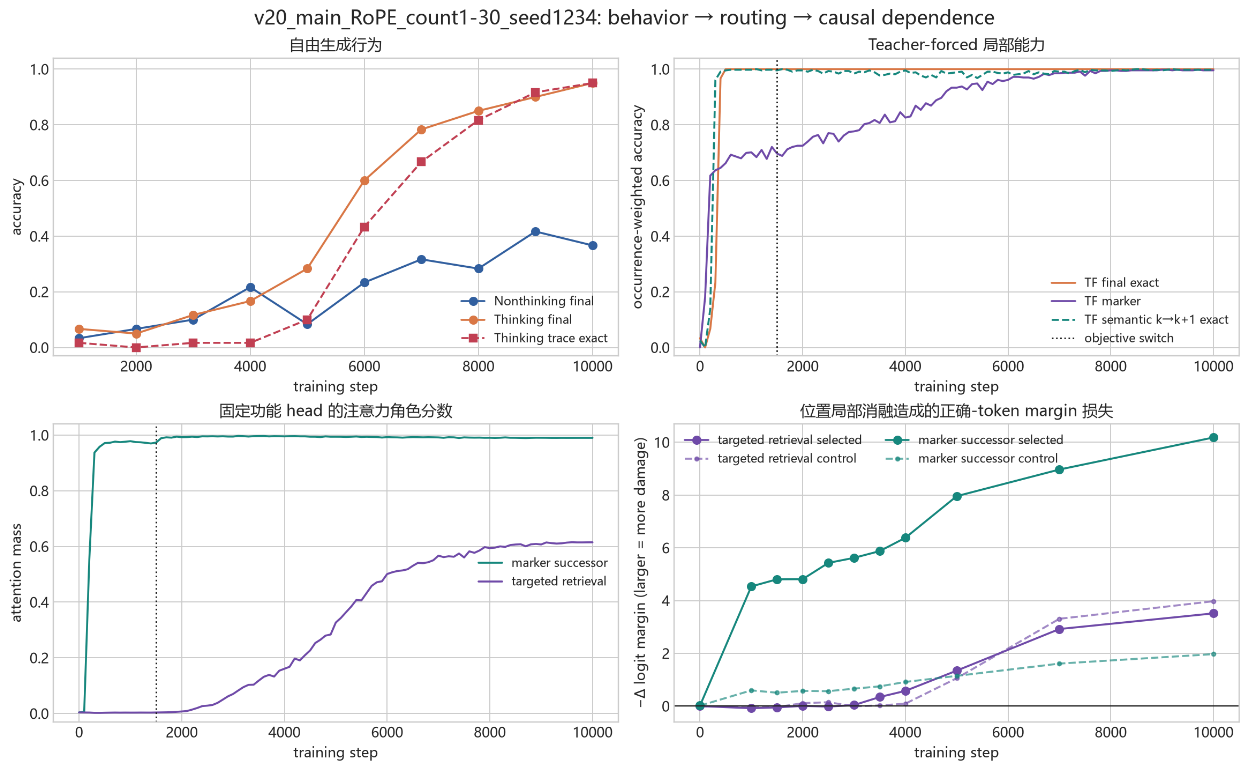

Figure 6: Current one-seed v20 training timeline. Teacher-forced local abilities appear before free-running trace accuracy; next-source behavior appears before matched-source specialization. Because the objective changes at step 1500, this ordering is descriptive until replicated under no-switch and multi-seed controls. Labels are inherited from the internal report.

6.4 Training-dynamics estimands

At saved checkpoints we track free-running exact accuracy, teacher-forced local accuracy, the three routing scores in equations (20), (21) and (23), count/progress SNR, query-local ablation damage, and clean patch recovery. Optimizer step is supplemented by semantic exposure: the cumulative number of supervised occurrences of each source identity, retrieval step, and answer count.

Three outcomes are distinguished:

1. **smooth formation**: continuous mechanism metrics predict behavior without a discrete change point;
2. **replicated transition**: a preregistered change-point model improves held-out likelihood and the ordering recurs across seeds and related causal metrics;
3. **objective/exposure artifact**: the apparent transition tracks the loss switch or event frequency rather than a mechanism-specific phase.

This follows work that uses mechanistic progress measures to explain apparently sudden learning [6, 7].

TODO. S1—multi-seed mechanism dynamics. Train five paired initialization seeds (1234/2234/3234/4234/5234) with a seed-independent frozen data pool and paired Direct/CoT batch stream. Save 100-step snapshots. Select each final-checkpoint functional bundle once on a discovery split, freeze it, and track the same bundle backward; do not reselect the highest-scoring head at each checkpoint. At preregistered milestones run query-local ablation and clean patching. Report each seed's 10–90% formation interval for next-source control, matched-source identity transport, progress update, final-total transport, and free-running exactness. Call an ordering replicated only if its sign agrees in at least four of five seeds and a paired interval excludes zero. Repair the failed causal stage and publish commands, manifests, and checksums.

TODO. S2—objective, exposure, and architecture controls. With at least three seeds per condition, vary one factor at a time: objective-switch step (0/1500/3000/10000), count sampling (natural/uniform), or context length (128/256/384). Plot formation against optimizer step, steps since the switch, and cumulative semantic exposure. Add a shared-backbone model with Direct/CoT mode tokens, loss-weight matching, digit-wise versus atomic count tokens, and held-out $C > 30$. If formation times track the switch or exposure, describe curriculum/exposure dynamics rather than an intrinsic phase transition.

7 Threats to Causal Identification

Retrieval versus aggregation failure. High list recall with low Direct count accuracy is compatible with aggregation or readout failure, but recall must be measured under the same prompt and decoding. Attention mass can dilute with length, and a multi-hop summary route can lack direct answer-to-source attention.

Visible-index and position leakage. Progress may be decoded from an item number, trace position, or answer distance. Primary anchors occur before source-bearing content or at item end, and analyses cross index permutation, index-free traces, fixed answer position, and variable trace extent.

Capability mismatch and floor effects. A clean CoT manifold and noisy Direct manifold may simply compare success with failure. We use all paired prompts, preregistered overlapping-difficulty cells, and four correctness strata. Common-correct analyses are explicitly selection-conditioned. If Gemma4-E4B Direct remains at floor, its role is failure-mechanism replication rather than unbiased mediation.

Different compute and decoding. Native CoT uses more generated tokens and may use a different sampling policy. Equal-length null scratchpads, fixed token budgets, deterministic matched decoding, and native-policy reruns separate semantic trace effects from extra compute and policy.

Post-treatment traces. Generated trace style is selected by the model. Canonical teacher forcing establishes local competence; free-running prefixes before the first error establish execution. Neither alone identifies a natural mediation effect.

Attention redundancy and superposition. One role may be distributed across heads, and one head may serve multiple roles. Bundle dose responses, union source coverage, query-local interventions, and clean recovery are preferred to naming one maximally scoring head.

Off-manifold interventions. Large centroid replacements can create unnatural states. Residual-preserving local paths and natural donor transplants are primary; activation norm, nearest-natural-state distance, and downstream KL are reported.

Cross-model non-homology. Qwen and Gemma may implement the same function with different depths and distributed components. Replication is defined by semantic endpoint and causal effect, not matching head indices.

8 Related Work

CoT prompting improves multi-step performance [12], but generated explanations can be unfaithful [9]. Recent work studies information flow and local causal structure in reasoning traces [11, 14]; counting adds exact source alignments and a known algorithmic target. Strong hidden-state probes can be diagnostic without being causal [16].

RULER extends single-needle retrieval to multi-needle and aggregation tasks [4]. Transformer counting can be limited by representation interference, vocabulary, and context length [15]. Our contribution is to compare the routing and state-update mechanisms of direct and trace-based counting in the same task.

The causal tools follow activation patching and causal tracing [5], circuit discovery [1], query-specific circuit analysis [10], and activation steering [8]. We distinguish local ordered-token successor primitives [3] from the next-source and continuation control required to cover a long-context source set.

9 Limitations

The causal LLM experiments in this draft are registered rather than complete. The existing behavioral audit mixes some checkpoint pairs, decoding policies, and protocol failure modes. Open-weight Qwen and Gemma results may not generalize to proprietary or non-autoregressive systems. Linear probes can miss nonlinear codes; flexible probes can memorize nuisances. Patching can be off-manifold, and ablation can trigger compensation. Native source alignment is imperfect after the first trace error. Synthetic models differ from LLMs in scale, pretraining, tokenization, and task familiarity. Finally, the response-surface forms were explored on existing data and must earn the name "law" by predicting the untouched Qwen3-14B cells.

10 Reproducibility and Artifact Plan

The final artifact will include request-level data or hashes, exact prompts, checkpoint/tokenizer revisions, chat templates, generation parameters, semantic-anchor parsers, intervention code, discovery/reporting splits, and seed-level tables. Each figure will be generated from a frozen table with a manifest and checksum. Numerical exactness, parse, format, and truncation remain separate. Synthetic checkpoints and optimizer states will be retained at the declared cadence.

TODO. R0—frozen analysis registry. Create one versioned YAML file containing: task unit; model/mode pairs; checkpoint hashes; core and mechanism cells; semantic sites; trace alignment rules; discovery/reporting groups; head scores and bundle selection; count-state and noise estimands; intervention dose grids; endpoint definitions; denominator gates; negative controls; multiplicity correction; minimum examples/seeds; exclusion codes; and the claim ladder in table 4. Generate every table caption's sample size and provenance tag from this registry.

11 Conclusion

The proposed answer to "How does CoT help language models count?" is a causal mechanism chain, not the observation that longer answers are better. Direct mode must make many relevant sources influence one late answer state through broad retrieval. CoT can instead retrieve one source at a time, preserve the result in a trace or internal progress state, and use next-source routing to continue until the source set is covered. This decomposition predicts lower source-identity and duplicate errors, a better-resolved final total state, and consequently higher exact accuracy.

The claim will be supported only if the same chain is recovered in both Qwen3-8B and Gemma4-E4B: targeted-route intervention must damage source identity or unique coverage, that damage must reduce final-state count discriminability and exact accuracy, and restoration of the clean intermediate or final state must rescue the answer while the route remains perturbed. If equal-length null scratchpads reproduce the gain, if attention changes without source or state changes, or if state restoration fails, the conclusion must be weakened to an extra-compute, routing-signature, or coverage-only account. The paper is structured so that such negative outcomes remain informative rather than being absorbed into an unfalsifiable narrative.

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A Complete Behavioral Tables and Model Exclusions

A.1 Requested response-surface panel on the full grid

Table 5: Preliminary numerical exact accuracy over the full $10 \times 5 \times 10$ grid (500 requests per available model–mode cell). These descriptive values precede the matched-policy rerun.

Model / logical slot	Direct	Index	Bullet	CoT
Qwen3-4B	.432	.888	.806	.928
Qwen3-8B	.570	.962	.894	.956
Qwen3-14B	*not yet evaluated*			
Qwen3-32B	.600	.974	.858	.976
Gemma4-E4B	.248	.882	.796	.832
GLM-4 / GLM-Z1†	.302	.710	.636	.684
Cogito-v1-Preview-8B	.376	.516	.776	.720
Nemotron-Nano-v2-9B	.236	.666	.580	.636
Nemotron-3-Nano-4B	.366	.652	.546	.654

†The GLM row stitches three modes from GLM-4-9B and CoT from GLM-Z1-9B; it is excluded from same-checkpoint mode effects. Gemma4-E4B Direct numerical exactness (.248) differs from registered success (.240); native CoT numerical exactness (.832) differs from registered success (.792).

A.2 Models excluded from the main panel

Models are not removed because they disagree with the hypothesis. They are excluded from the main response-surface or mechanism panel when a frozen evaluation gate fails: incomplete mode coverage, a different checkpoint for the reasoning condition, severe parse/format/truncation confounding, or accuracy too close to floor for the planned causal contrast. Their numerical accuracy remains visible.

Table 6: Other audited models and why they are assigned to the appendix. Entries are full-grid numerical exact accuracy; a dash indicates that the mode was not run on that checkpoint.

Model / checkpoint setup	Direct	Index	Bullet	CoT	Main-panel exclusion reason
Qwen3-1.7B	.196	.288	.396	.602	Low-capability floor; enumeration also has substantial parse/truncation failures. Retained as a phase-boundary sensitivity.
Gemma4-12B	.514	.950	.746	.908	Native numerical accuracy is high but registered success is .314 because of final-channel format behavior. Not selected for the co-primary 8B-scale mechanism comparison.
DeepSeek-R1-0528-Qwen3-8B	–	–	–	.672	Native-only cell; no within-checkpoint mode quartet.
Granite-3.3-Instruct-8B	.394	.440	.374	.440	Low and weakly separated modes plus format/protocol failures; unsuitable for non-floor mechanism mediation.
Ministral Instruct / Reasoning	.658	.892	.756	.770	CoT uses a separate reasoning checkpoint; native registered success is .020 despite .770 numerical exactness, creating a severe format confound.
OLMo3 Instruct / Think	.234	.238	.302	.508	CoT uses a separate checkpoint; Native parse is .746 and truncation .254, while Direct is near floor.

The two Nemotron models remain in the requested cross-family panel, but their full per-model Index/Bullet surfaces are appendix material because enumeration has nontrivial parse/truncation failures. Nemotron-3 Native is comparatively clean. Cogito Index likewise requires parser/protocol repair before conditional signed-bias fits can be interpreted.

B Registered Experiment Ledger

Table 7: Dependencies and claim gates for all TODO experiments.

ID	Target claim	Frozen primary test	Claim gate
B0	Paired behavior to explain	500 contexts, four modes, matched/native decoding, token-budget cross, Qwen3-8B and Gemma4-E4B	Paired numerical gap survives protocol decomposition and is not entirely extra compute
B1	Quantitative N, L/size law	Fit only Qwen 4B/8B/32B; predict untouched 14B 2,000-request grid	Better held-out prediction than no-size baseline with stable form; otherwise qualitative surfaces
B2	Protocol/checkpoint validity	GLM four-mode completion; Gemma Direct and Cogito Index repair	Same-weight comparisons and acceptable parser/selection sensitivity
R1	Count/progress representations	Dense 4k plus sparse 8k; grouped holdout; full-space spacing and SNR	Count information generalizes beyond position/token controls; final sites compared fairly
R2	Causally executable state	Local curve steering and natural-state transplants with bidirectional/random controls	Complete parsed count moves monotonically and natural patches agree
R3	Iterative counter	New-valid/invalid/duplicate event triples in prompt and trace	+1, no-op, persistence, generalization, and downstream transport
H1-D	Direct broad-source route	Multi-source edits; query-local bundle ablation/restoration at q_D	Multiple sources, Z_D, and parsed answer are damaged and recovered
H1-C	CoT matched-source route	Same-position identity swaps; pattern/value/head/residual interventions	Source state and emitted identity recover; bundle retargets semantically
H2	Next-source/continue-stop route	Nested short/long prefixes; bidirectional item-end patching	Next source, continue/close, and rollout stopping change in both directions
M1	Retrieval-to-state-to-answer chain	Target-head × identity-state $\times Z_C$ factorial intervention	Ordered loss and rescue of identity, coverage, state quality, and exact count
M2	Coverage versus aggregation noise	Fixed-U teacher forcing; faithful/corrupt/null/prefilled trace controls	Fixed-coverage Z_C effect required for a beyond-coverage noise claim
M3	Cross-model explanation of gap	Frozen endpoints, independent Gemma discovery, 300 free-running traces/model	Gap removal and restoration replicate in both models before first error
S1	Synthetic formation order	Five paired seeds, 100-step snapshots, frozen final bundle tracked backward	Ordering agrees in at least four seeds with paired interval excluding zero
S2	Dynamics not an objective artifact	Switch-step, exposure, length, shared-backbone, and tokenization controls	Timing is not explained by switch/exposure before any phase-transition claim
R0	Reproducible analysis registry	Versioned YAML plus manifests/checksums	All figures and tables reconstruct from frozen inputs and declared exclusions

C Prompt, Trace, and Intervention Controls

- Direct total-only; numbered list; bullets; index-free item tags; native CoT; equal-length semantically null scratchpad.
- Same faithful list in user versus assistant prefix; model-generated list replayed under teacher forcing; Direct with a prefilled ledger.
- Correct identity with wrong attribute; wrong identity with correct attribute; duplicate, missing, invalid, and same-template distractor records.
- Candidate ordinal fixed while validity flips; number of seen candidates fixed while valid-prefix count changes.
- Source order permutation, visible-index permutation, symbolic indices, and item anchors before versus after visible numerals.

- Fixed trace extent with different unique coverage; fixed coverage with different duplicate counts; fixed answer position with variable trace extent.
- Fixed (N, L) with dispersed versus clustered targets and early/middle/late source-depth strata.
- Same final total with different source sets and traces; same trace length with different totals.
- Query-local mean ablation, zero ablation sensitivity, score-matched/random/wrong-layer head bundles, wrong query rows, wrong source donors, and norm-matched subspaces.
- Complete parsed numeric-string probability and free-running exact count, never only a first-digit logit.

D Additional Statistical Specification for Representation Noise

For mode m, site s, layer ℓ, and count c, fit a grouped cross-validated decoder on training groups only. On held-out groups define decoder bias, variance, and mean squared error:

$$b_{ms\ell}(c) = \mathbb{E}[\widehat{C} - C \mid c], \qquad v_{ms\ell}(c) = \mathrm{Var}(\widehat{C} - C \mid c), \qquad q_{ms\ell}(c) = v_{ms\ell}(c) + b_{ms\ell}(c)^2. \tag{34}$$

Using the centroids and pooled covariance in equation (16), normal-direction normalized dispersion and shrinkage Mahalanobis separation are

$$\nu_{\perp,j}^{\ell} = \frac{\mathrm{tr}\Big[(I - t_j t_j^{\top})\overline{\Sigma}_j^{\ell}\Big]}{(d_{\mathrm{model}} - 1)\big((d_j^{\ell})^2 + \epsilon\big)}, \tag{35}$$

$$D_{M,j}^{2,\ell} = \frac{\big(\mu(c_{j+1}) - \mu(c_j)\big)^{\top}\big(\overline{\Sigma}_j^{\ell} + \lambda I\big)^{-1}\big(\mu(c_{j+1}) - \mu(c_j)\big)}{(c_{j+1} - c_j)^2}. \tag{36}$$

Tangential dispersion is primary because it directly confuses adjacent count coordinates; normal dispersion detects off-curve heterogeneity. All covariance estimators and λ are selected within the fit split.

On the dense $C = 1{:}30$ mechanism set, estimate prespecified slopes of $\log(\nu_{\mathrm{state}} + \epsilon)$, $\log(v + \epsilon)$, and $\log(q + \epsilon)$ against count while controlling for context length, target-depth distribution, answer position, trace extent, and correctness stratum. Sparse-grid analyses regress on interval midpoint and use the observed Δc; they never pretend that an unobserved adjacent count exists. Cluster bootstrap intervals resample documents and context seeds. The main comparison is Direct Z_D versus CoT Z_C, with Index separate. A larger-count-noise claim requires convergent spacing, dispersion, decoder-error, and causal-transport evidence rather than one significant slope.

E Architecture-Specific Intervention Notes

Both co-primary models use grouped-query attention, so an intervention must specify whether it acts on a query head, a shared key/value head, or the concatenated pre-output-projection head slice. For each layer, the artifact records tensor shape, rotary position convention, attention backend, and hook location. The causal decomposition is:

$$\begin{aligned}&\text{attention pattern} \to \text{source-weighted value} \to \text{query-head output before } o_{\mathrm{proj}}\\ &\qquad \to \text{attention block output} \to \text{post-layer residual.}\end{aligned}$$

Pattern-only recovery supports routing; value/head-output recovery supports information transport; residual-only recovery localizes the effect no more finely than the layer. MLP output is patched separately only after a block-level state effect is established. Cross-mode and cross-model activation patches are exploratory because the residual bases and manifolds need not align.

F Claim-to-Evidence Reporting Rules

1. Every exact number in the main text must trace to a frozen CSV/table and a manifest, not only to HTML prose.
2. "Counter curve" denotes full-space ordered centroids; PCA is a projection. "Iterative counter" additionally requires event-level update/no-op and downstream transport.
3. "Broad," "matched-source," and "next-source" are candidate labels until independent query-local ablation and clean restoration establish the causal endpoint.
4. Teacher-forced effects establish local competence. Natural execution requires the same signed effect on free-running prefixes before the first error.
5. Synthetic results establish measurement feasibility, not the mechanism of a pretrained LLM.
6. A response surface becomes a law only after untouched-model prediction. Failed validation remains in the paper as a negative result.